VISION VIGILANTS | SMART INDIA HACKATHON 2026

Technology Stack Report

IBVAP - Intelligent Border Video Analytics Platform

Field	Value
Document	Technology Stack Report
System	IBVAP - Intelligent Border Video Analytics Platform
Team	Vision Vigilants
Compiled	11 September 2026
Supersedes	docs/IBVAP-Technical-Report.pdf (9 September 2026)
Basis	Read directly from the working installation, not from the requirements file. Where a pinned range and the installed version differ, the installed version is given.

How to read this document

Every version number, count and measurement here was taken from the running project on 11 September 2026. Nothing is estimated. Section 13 lists what the system does *not* do and where it is weak, because a stack report that only lists strengths is not useful to anybody deciding whether to trust it.

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1. System at a glance

IBVAP watches a video source, decides what it is looking at, decides whether that matters, and records the decision. Four layers do that work, and each one only talks to the next.

Layer	Responsibility	Built with
Perception	Find objects in a frame and keep the same object's identity across frames	YOLOv8n, YOLOv8n-pose, ByteTrack
Analysis	Turn positions over time into behaviour, boundary state and an anomaly score	Own Python modules, scikit-learn Isolation Forest
Judgement	Score the threat, raise an incident, raise an alert, capture evidence	Own rules engine driven by threat_rules.yaml
Presentation	Serve the live feed, the operator dashboard and the recorded history	FastAPI, React, SQLite

Measure	Value
Python source	68 files, 25,063 lines (core, api, database, inputs, app, tools, scripts)
Frontend source	12 files, 21,560 lines
Automated tests	460, all passing
Analysis modules	29 in core/
HTTP endpoints	27 (18 surveillance, 9 authentication and admin)
Database tables	16
Model weights	3 files, 19.6 MB total
Runtime processes	1 (FastAPI serves the compiled dashboard itself)

2. Runtime and platform

Component	Version	Note
Python	3.14.3	Homebrew build, virtualenv at .venv
Platform	macOS on Apple Silicon (arm64)	Development and demo machine
Acceleration	PyTorch MPS when available	detection.device is "auto"; falls back to CPU without changing behaviour
Process model	Single process	One uvicorn serving both the API and the compiled dashboard on port 8000

> **Why one process and not two**
>
> Until recently the system ran uvicorn for the API and the Vite dev server for the interface, on two ports. Two processes means two things that can be stale, and a uvicorn left running from before an endpoint existed produced two separate reports of the application being broken which were really the old server still answering. FastAPI now serves the built frontend itself. One process cannot disagree with itself, and CORS is gone with it.

3. Perception: models and detection

3.1 Model weights

File	Size	Purpose
yolov8n.pt	6.5 MB	Object detection. COCO-pretrained nano variant
yolov8n-pose.pt	6.8 MB	Human pose, 17 COCO keypoints
border_yolo.pt	6.2 MB	Project detection weights
behaviour_anomaly.joblib	1.1 MB	Isolation Forest, trained on this deployment's own movement features

3.2 Detection settings

Setting	Value	Why
Confidence	0.35	Below this a detection is discarded
IoU	0.50	Non-maximum suppression overlap
Image size	640 px	Inference resolution
Frame stride	2	Every second frame is analysed. Halves the compute and does not lose events, because a person cannot cross a boundary inside one frame
Classes	person, car, motorcycle, bus, truck, bicycle	Six of the eighty COCO classes. The rest are noise for this task

3.3 Tracking

Setting	Value	Why
Tracker	ByteTrack	Supplied by Ultralytics, run with persist=True
Minimum hits	3	A track is not published until it has been seen three times, which removes single-frame false positives
Coast frames	5	How long a track survives without a detection before it stops being published
Max age	30 frames	How long its identity is held for re-identification after that
Trajectory	240 points	Path history retained per track

3.4 Pose

Setting	Value	Why
Model	YOLOv8n-pose	17 COCO keypoints
Crop size	192 px	Pose runs on a crop of the person, not the whole frame, so a distant figure still gets usable resolution
Interval	every 2nd frame	Pose is the expensive stage
Minimum height	0.12 of frame	Below this the joints are not separable and the result would be invented
Maximum persons	4 per frame	Bounds worst-case cost per frame
Keypoint threshold	0.30	Below this a joint is treated as unseen

4. The analysis chain

Pixels become a judgement through nine stages. Each is a separate module in `core/` and each can be tested on its own, which is why the test suite can cover behaviour classification without running a detector.

#	Stage	Module	What it produces
1	Detect	detector.py	Boxes and classes for one frame
2	Track	tracker.py	A stable identity per object over time
3	Pose	pose.py, pose_tracker.py	17 keypoints per person
4	History	pose_history.py	A time series of poses per track
5	Movement	movement.py	Speed, direction, posture, all measured in the person's own body heights so the result does not depend on camera distance
6	Behaviour	behaviour_classifier.py	WALKING, RUNNING, BENT_WALK, BENT_RUN, CROUCHING, JUMPING, FALLING, STANDING
7	Boundary	boundary_interaction.py	NONE to APPROACHING to AT_BOUNDARY to INTERACTING to CROSSED
8	Anomaly	anomaly_engine.py	An Isolation Forest score against what this camera normally sees
9	Threat	threat_engine.py	A weighted score and a level

4.1 Threat scoring

The threat engine adds weighted factors. It is deliberately additive and deliberately in a config file, so the reasoning can be read and argued with rather than being buried in code. Sixteen factors are defined in `config/threat_rules.yaml`, including restricted zone intrusion, boundary crossing, behaviour, loitering duration, behavioural anomaly, watchlist match, night time and group presence.

Level	Score	Meaning
LOW	0 to 24	Recorded, not alerted on by default
MEDIUM	25 to 49	Alert raised, evidence captured
HIGH	50 to 74	Alert raised, evidence captured
CRITICAL	75 and above	Alert raised, evidence captured

> **Scale independence**
>
> Every distance and speed in the movement stage is measured in the detected person's own body heights, not in pixels. A person walking scores about 0.45 and running about 1.9 whether they are close to the camera or far from it. This is what allows one set of thresholds to work across four cameras with different framing, and it is the reason the system needs no per-camera calibration.

5. ANPR and registration decoding

Number plate recognition runs only inside a detected vehicle box, so it costs nothing on a camera with no vehicles in it. It is optional: with no OCR engine installed the system reports "No OCR backend installed" and every other capability continues unaffected.

5.1 Pipeline

Step	Method
Region proposal	Blackhat morphology to raise dark-on-light text, a Sobel pass for the dense vertical edges characters make, then a closing operation to merge them into one blob per plate. Candidates filtered on aspect ratio and relative area
OCR	EasyOCR preferred, Tesseract as fallback
Validation	Regex for the Indian plate shape: two letters, one or two digits, up to three letters, three or four digits
Consolidation	Reads are grouped by tracked vehicle and resolved by a confidence-weighted vote, so one car produces one answer rather than one row per frame
Origin	State and district decoded from the registration prefix

5.2 Engine comparison, measured on the project's own footage

Both engines were run through the project's own ANPR module over the CAM-04 highway clip, sampling every tenth frame: 49 frames, 169 vehicle boxes above the minimum height.

Engine	Reads	Hit rate	Best result
EasyOCR	23	13.6%	PB39N0203 at 0.99 confidence, format valid
Tesseract	3	1.8%	One plate found in the entire clip

5.3 RTO origin decoder

A new module, `core/rto.py`, maps a plate prefix to the state and district that issued it. PB39 is Rajpura in Punjab; HR26 is Gurgaon in Haryana.

Property	Value
State codes	41, covering all 36 states and union territories plus the BH nationwide series and the pre-rename OR and UA spellings
District codes	998
Dependencies	None beyond the standard library
Decoded at	Read time, not capture time
Misread repair	Letter-for-digit confusions are corrected in the district position only, where an Indian plate carries digits and nothing else. HPO1 becomes HP01. The series letters after it are never touched
Tests	27

> **Why the origin is not stored**
>
> The state and district are a pure function of the plate string, so storing them would freeze today's answer into every row. Decoding when a record is served means correcting or extending the table immediately improves every plate already captured, and it meant the capture pipeline did not have to change at all to gain the capability.

What the district table does not cover

The public register the table was compiled from predates Telangana (2014) and Ladakh (2019), so those decode to the state with no district. The module reports that explicitly as "district not in table" rather than returning a blank that would look like a failed read.

6. Backend and API surface

Package	Version	Role
FastAPI	0.141.1	HTTP framework and request validation
Starlette	1.6.0	ASGI layer beneath FastAPI
Uvicorn	0.52.4	ASGI server
Pydantic	2.13.5	Request and response models
httpx	0.28.1	Used by the test client

6.1 Surveillance endpoints

Method	Path	Returns
GET	/	The compiled dashboard, or a status document if it has not been built
GET	/api/health	Liveness. Open, so the login screen can report reachability before anyone has a token
GET	/api/cameras	Configured cameras and their status
GET	/api/cameras/{id}/stats	Live counters for one camera
GET	/api/cameras/{id}/tracks	Current tracks with their analysis
GET	/api/cameras/{id}/stream	Annotated MJPEG feed
GET	/api/cameras/{id}/events	One camera's recorded events
POST	/api/cameras/{id}/start	Start analysis
POST	/api/cameras/{id}/stop	Stop analysis
GET	/api/dashboard	Fleet totals
GET	/api/events	Recorded events, filterable by camera and session
GET	/api/incidents	Raised incidents
GET	/api/alerts	Raised alerts
GET	/api/evidence	Captured snapshots with their incident caption
GET	/api/evidence/{id}/image	One evidence JPEG
GET	/api/plates	Every individual OCR attempt
GET	/api/plates/{id}/image	One plate crop
GET	/api/vehicles	One row per vehicle, plate resolved and decoded

6.2 Authentication endpoints

Method	Path	Access
POST	/api/auth/login	Open
POST	/api/auth/logout	Any signed-in user
GET	/api/auth/me	Any signed-in user
GET	/api/admin/users	Super Admin
POST	/api/admin/users	Super Admin
PUT	/api/admin/users/{id}	Super Admin
DELETE	/api/admin/users/{id}	Super Admin
POST	/api/admin/users/{id}/unlock	Super Admin
GET	/api/admin/audit-logs	Super Admin

> **Why two routes accept a token in the query string**
>
> A browser will not attach an Authorization header to an `<img>` request, and the camera stream and the evidence images are both loaded that way. The choice is a query token or an unauthenticated video feed, and an unauthenticated feed would hand anybody the cameras. The cost is that the token appears in the server access log, which is why no other route accepts one this way.

7. Data layer

Property	Value
Engine	SQLite, standard library, WAL journal mode
Location	data/database/borderguard.db
Tables	16
Migrations	schema_version table
Server needed	None. The database is a file, which is what makes the whole system runnable from one double-click

7.1 Tables

Group	Tables	Holds
Configuration	cameras, zones, settings, schema_version	What is configured and which schema version it is
Runtime record	sessions, tracks, detections	One row per run, per tracked object, per sampled detection
Judgement	incidents, alerts	What the analysis concluded, and the sentence the operator was shown
Evidence	evidence, plates	Captured images indexed to the incident that caused them
Security	users, auth_sessions, audit_logs	Accounts, live sessions and who did what
Other	watchlist, system_logs	Flagged plates and diagnostics

> **Evidence is files plus an index, not blobs**
>
> Images are written to data/evidence/ by date and indexed in the evidence table with their incident, camera, track, size and timestamp. The database stays small and fast, the images stay inspectable with any image viewer, and retention can delete by date without rewriting the database. An image is served only through an authenticated route, never by path.

8. Authentication and authorisation

Two roles: SUPER ADMIN and NORMAL USER. Authorisation is enforced by the backend on every request. The React interface hides pages a role may not use, but that is a convenience, not the control: the endpoints refuse the role independently and record the refusal.

Property	Value	Note
Password hashing	scrypt	hashlib, standard library. No new dependency was added for security
Cost parameters	N = 32768, r = 8, p = 1	64-byte derived key
Salt	16 bytes, per password	Stored in a self-describing hash string
Comparison	hmac.compare_digest	Constant time
Session token	secrets.token_urlsafe(32)	Opaque and random
Token at rest	SHA-256 fingerprint only	The database never holds a usable token, so a database copy cannot be replayed
Idle timeout	30 minutes	Warning shown 120 seconds before expiry
Absolute timeout	12 hours	Regardless of activity
Lockout	5 failed attempts, 15 minutes	Per account
Minimum password	8 characters	Maximum 256

Two decisions worth defending

First, the same message is returned for an unknown user and a wrong password, and a dummy hash verification runs on the unknown-user path so the two take the same time. An attacker cannot use the login form to discover which User IDs exist. Second, no password is stored in config/application.yaml. A password in config is a password in Git, and deleting it later does not remove it from the history. The first admin is created by scripts/create_admin.py, which prompts without echoing and never writes the plain text anywhere.

9. Frontend

Package	Version	Role
React	19.2.8	Interface
React DOM	19.2.8	Rendering
Vite	8.2.2	Build tool
@vitejs/plugin-react	6.1.1	React support in Vite
oxlint	1.81.0	Linting

Two runtime dependencies. No UI framework, no component library, no state management library, no router, no charting library. The interface is hand-written React with hand-written CSS. That is a deliberate choice: it keeps the production bundle at 282 KB of JavaScript and 53 KB of CSS, and it means nothing in the interface is a black box during a demonstration.

Property	Value
Production bundle	282 KB JavaScript, 53 KB CSS (84 KB and 11 KB gzipped)
Served by	FastAPI, from frontend/dist, same origin
API base	Empty in production, absolute in development. Decided at runtime from the port, so one bundle works in both and the production build cannot drift from the development build
Pages	Dashboard, Live Monitoring, Camera Management, Alerts, Detection Logs, Analytics, User Management, Audit Logs, System Settings

9.1 Detection overlays

Bounding boxes, person numbers, behaviour labels and threat levels are drawn into the frame by `core/annotator.py` before it leaves the backend, not by React. What is on screen is therefore exactly what the analysis saw. The interface cannot draw a box the pipeline did not produce, which is the point.

10. Packaging and launch

Component	Detail
Launcher	IBVAP.app, a macOS bundle. Double-click to start
Logic	scripts/launch_ibvap.sh, version-controlled and editable, rather than buried inside the bundle
Stale server	Any running IBVAP server is stopped before a new one starts, every time
Readiness	The launcher polls /api/health and confirms the new process owns the port before opening a browser, instead of guessing with a fixed sleep
Failure	Reported in a macOS dialog and written to logs/launcher.log, because a double-clicked app has nowhere to print
Shutdown	Quitting from the Dock stops the server, so the next launch starts clean
Portability	The venv interpreter is invoked directly rather than sourcing activate, so moving the project folder cannot break it

> **Why not a single PyInstaller binary**
>
> Bundling PyTorch, Ultralytics and OpenCV into one executable produces a 2 to 4 GB artifact, takes a long time to build, and routinely breaks model-weight paths and hardware acceleration detection. The launcher approach gives the same single double-click with none of that fragility.

11. Testing

460 tests, all passing, run with pytest 8.4.2. They are unit tests over the analysis modules rather than screenshots of the interface, which is what allows behaviour classification and threat scoring to be checked without running a detector.

Test file	Covers
test_activity.py	Movement measurement and the activity chain
test_anomaly_threat_incidents.py	Anomaly scoring, threat rules, incident deduplication
test_auth.py	Hashing, sessions, lockout, role enforcement
test_behaviour_alerts.py	Which behaviours raise their own incident
test_boundary_interaction.py	The five boundary states and crossing confirmation
test_database_config_inputs.py	Schema, config loading, video sources
test_end_to_end.py	The real detector over real footage
test_event_log.py	Event scoping, so one camera's events cannot appear under another
test_geometry_zones_lines.py	Polygon and line geometry
test_inference_gate.py	Frame skipping and inference budgeting
test_low_light.py	Night-time handling
test_pose.py	Keypoint extraction and pose history
test_rto.py	Plate origin decoding and its refusal to invent
test_tracking_behaviour.py	Track identity and behaviour over time
test_zone_optionality.py	A camera with no zones must still work

12. Full dependency inventory

Installed versions, read from the virtual environment on 11 September 2026. Where requirements.txt pins a range, the version actually in use is given here.

12.1 Machine learning and computer vision

Package	Version	Role
torch	2.14.0	Tensor and neural network runtime
torchvision	0.29.0	Vision operators and transforms
ultralytics	8.4.142	YOLO models plus the ByteTrack tracker
ultralytics-thop	2.1.6	Model complexity accounting
opencv-python	4.14.0.94	Frame capture, image operations, drawing
opencv-contrib-python	5.0.0.93	Additional OpenCV modules
mediapipe	1.0.1	Additional pose tooling
scikit-learn	1.9.0	Isolation Forest for behavioural anomaly
scipy	1.18.1	Numerical routines
numpy	2.5.3	Arrays, used everywhere
lap	0.5.13	Linear assignment, used by ByteTrack
joblib	1.6.0	Persisting the anomaly model
sympy	1.14.0	PyTorch dependency
networkx	3.6.1	PyTorch dependency

12.2 Web and API

Package	Version	Role
fastapi	0.141.1	HTTP framework
starlette	1.6.0	ASGI foundation
uvicorn	0.52.4	ASGI server
pydantic	2.13.5	Validation and serialisation
pydantic-core	2.46.5	Pydantic's compiled core
httpx	0.28.1	HTTP client, used by tests
httpcore	1.0.9	httpx transport
h11	0.16.0	HTTP/1.1 protocol
anyio	4.15.1	Async compatibility layer
requests	2.34.2	HTTP client
urllib3	2.7.0	HTTP plumbing
click	8.5.0	Command line parsing for uvicorn

12.3 Data, configuration and desktop

Package	Version	Role
pandas	2.3.3	Tabular analysis

Package	Version	Role
polars	1.44.1	Fast dataframe operations
pyyaml	6.0.3	All four configuration files
matplotlib	3.11.1	Plotting for analysis tooling
pillow	12.3.0	Image handling
PySide6	6.11.2	Qt desktop interface (alternative to the browser)
shiboken6	6.11.2	PySide6 binding layer
psutil	7.2.2	Process and resource inspection
sounddevice	0.5.6	Audio alerting
pytest	8.4.2	Test runner

12.4 Optional: number plate recognition

Package	Status	Note
easyocr	Recommended, not yet installed	Best accuracy on this footage by a factor of seven. Adds about 100 MB of weights on first run
pytesseract	Fallback, not installed	Also needs the tesseract system binary

> **Current state of ANPR**
>
> Neither OCR engine is installed on the demonstration machine, so ANPR is switched off and the dashboard reports ANPR UNAVAILABLE with the command to enable it. Detection, tracking, behaviour and threat analysis are unaffected. This is the intended behaviour for an optional capability: it declares its absence rather than silently returning nothing.

12.5 JavaScript

Package	Version	Type
react	19.2.8	Runtime dependency
react-dom	19.2.8	Runtime dependency
vite	8.2.2	Build only
@vitejs/plugin-react	6.1.1	Build only
oxlint	1.81.0	Build only
@types/react	19.2.18	Build only
@types/react-dom	19.2.4	Build only

13. Limitations, stated plainly

A stack report that lists only strengths tells a reader nothing they can act on. Everything below is a real limit of the system as it stands on 11 September 2026, and each was found by measuring rather than by guessing.

13.1 Capability

Limit	Detail
ANPR accuracy	On the highway clip EasyOCR reads 5 plates from 20 tracked vehicles. The rest are too distant, too angled, or in traffic where a plate is a few pixels wide. Ground truth on one vehicle is HR26BQ0008; the best read was HR260000081, with BQ misread as 00. The state and district still decode correctly because those come from the first four characters
Two behaviours unexercised	The classifier defines CROUCHING and JUMPING and scores them correctly, but neither occurs in any of the project's own footage. The rules are right and untested against real video
Not implemented	Weapon detection, face recognition and watchlist OCR matching are out of scope and are not claimed
District coverage	998 district codes, which is not every code in India. Telangana and Ladakh decode to the state only. The module says so rather than guessing
No calibration	Distances are in body heights, not metres. The system cannot report that somebody moved four metres, only that they moved two body heights

13.2 Footage

Source	Status
CAM-01 fence_own.mp4	Filmed by the team. Cleared for submission
CAM-02 img_3894_eval.mp4	A phone recording of a screen playing film footage. Development input only, not cleared
CAM-03	Stock footage carrying a visible watermark. Development input only, not cleared
CAM-04 ladakh_convoy.mp4	Dashcam footage with real Indian plates. Used for ANPR evaluation

13.3 Operational

Limit	Detail
Single machine	One process, one SQLite file, one host. There is no clustering and none is claimed
Stream token in logs	The MJPEG and evidence routes accept a session token in the query string, which the access log records. Accepted deliberately, because a browser cannot send a header on an image request
macOS bundle unsigned	IBVAP.app has no Apple Developer signature, so Gatekeeper blocks the first launch until the user approves it once
CPU inference	Without MPS or CUDA, expect roughly 60 to 120 ms per frame with yolov8n

> **The principle behind all of the above**
>
> The system is built so that it cannot show a number it did not compute. An empty panel says which of three things happened: the engine is not installed, nothing has been detected yet, or the backend could not be reached. A plate that could not be read says so rather than showing the least-bad guess. A district that is not in the table is reported as missing rather than left blank. This is the reason the limitations above can be listed at all: the system reports them itself.

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